

Fraud Detection System using Machine Learning

Abstract

This project presents the design and implementation of a fraud detection system for credit card transactions. The solution integrates machine learning models, data preprocessing, and deployment as a Streamlit web application. Due to anonymized real-world datasets (features V1–V28), a simulated dataset with interpretable features was used, making the project user-friendly while preserving realistic fraud detection challenges.

Introduction

Financial fraud is a growing concern as online and credit card transactions continue to rise. Detecting such frauds requires handling class imbalance, building efficient models, and ensuring interpretability. This project demonstrates the end-to-end machine learning pipeline: from exploratory data analysis (EDA) and feature engineering to training robust models and deploying them into a functional application.

Tools Used

- Python – core programming language
- Pandas & NumPy – dataset manipulation and numerical operations
- Scikit-learn – preprocessing, scaling, evaluation metrics
- XGBoost – classification model for fraud detection
- Seaborn & Matplotlib – visualization and analysis
- Streamlit – web application deployment

Steps Involved in Building the Project

- Data Exploration & Cleaning – performed EDA, handled noisy data, and ensured feature clarity.
- Feature Engineering – encoded categorical features and standardized numeric ones.
- Balancing Strategy – identified heavy class imbalance, evaluated outlier detection models.
- Model Training – applied Isolation Forest, Local Outlier Factor, and XGBoost; XGBoost selected for accuracy and interpretability.
- Evaluation – metrics included confusion matrix, ROC-AUC, precision, recall, and F1-score.
- Visualization – developed plots showing fraud vs normal distribution, feature impact, and categorical breakdowns.
- Deployment – implemented a Streamlit app with two modules: prediction (single/bulk transactions) and visualization (EDA insights).

Conclusion

The fraud detection project highlights the importance of robust preprocessing, model selection, and practical deployment. By moving from raw data to a deployed application, the project demonstrates the real-world workflow of AI/ML development. Using a simulated dataset allowed greater interpretability, while the deployment provided an accessible interface for fraud analysis.